

Abasyn University Islamabad Campus

Course Specification Form

1. Department	2. Program
Computing	BS Artificial Intelligence

3. Course Code	4. Course Title	5. Lectures Timing & Location	
AI 216	Programming for Artificial Intelligence	Monday	08:30 – 10:30
6. Credits	7. Pre-requisites	8. Course Web-Page	9. Course Coordinator
2+1	-	Lms.abasynisb.edu.pk	

10. Academic Year	11. Semester
2025-2026	Spring 2026

12. Course Instructor	13. Email	14. Office	15. Office Hours
Asad Hanif	asad.hanif@abasynisb.edu.pk		

16. Textbook(s)
1. Severance, C.R., 2016. "Python for everybody: Exploring data using Python 3." CreateSpace Independent Publ Platform. 2. Miller, B.N., Ranum, D.L. and Anderson, J., 2019. "Python programming in context." Jones & Bartlett Pub. 3. McKinney, W., 2012. "Python for data analysis: Data wrangling with Pandas, NumPy, and IPython." O'Reilly Media, Inc.
17. Reference(s)
1. Russell, S. and Norvig, P. "Artificial Intelligence. A Modern Approach", 3rd ed, Prentice Hall, Inc., 2015. 2. Joshi, P., 2017. "Artificial intelligence with python." Packt Publishing Ltd. 3. Janert, P.K., 2010. "Data analysis with open source tools: a hands-on guide for programmers and data scientists." O'Reilly Media, Inc.
18. Other Resources used (e.g. e-learning, field visits, periodicals, software, etc.):
https://www.w3schools.com/python/python_intro.asp https://pll.harvard.edu/course/cs50s-introduction-artificial-intelligence-python https://www.coursera.org/learn/ai-python-for-beginners
19. Course Description (as approved by the dept.)
The first objective of the course is to introduce and then build the proficiency of students in different AI programming languages. The basics include IDE for the languages, variables, expressions, operands and operators, loops, control structures, debugging, error messages, functions, strings, lists, object-oriented constructs and basic graphics in the languages. Special emphasis is given to writing production quality clean code in the programming language. Once the classical programming languages are properly introduced, the course should introduce some libraries necessary for interpreting, analyzing and plotting numerical data in Python (e.g., NumPy, Matplotlib, Anaconda and Pandas for Python) and give examples of each library using simple use cases and small case studies

20. Course Learning Outcomes (CLOs)			
CLOs	Mapping to PLOs		
On successful completion of the course, students will be able to:	Domain	Taxonomy Level	PLO

1) Understand the fundamental constructs of Python programming languages.	Cognitive	2	
2) Comprehend the fundamental constructs of programming languages for data analysis and representation.	Cognitive	2	
3) Understand and apply the Object-oriented concepts in the programming languages.	Cognitive	2	
4) Apply various libraries for plotting, interpreting and analyzing data in Python.	Cognitive	3	

21. Course Assessment		
Assessment Type	Weight	Date(s)
Midterm	25	To be announced by examination
Quizzes	10	2 Before Midterm, 2 Before Final Exam
Assignments	10	Due dates as mentioned on the assignments
Final Exam	50	To be announced by examination
Project	05	Before Final Exam
Total	100	

22. Course Teaching Methods
1) Interactive learning
2) Problem solving learning
3) Practical and laboratory learning
4) Self-directed learning
5) Group work and project

23. Course Weekly Breakdown			
Week	Date	Topics Covered	Lab Activities
1		Intro to course & GitHub setup	
2		Intro to python programming	
3		Intro to object oriented programming with Python	
4		Intro to data structures in Python programming	
5		Intro to Numpy and Pandas	
6		Data collection through web scrapping and database	
7		Reading data from different files & Data preprocessing	
8		Exploratory data analysis tools (plots, graphs, statistics)	

9		Mid Term	
10		Exploratory data analysis: A case study	
11		Basic text processing (BOW, stemming, regular expressions)	NLP/CV
12		Decision Tree implementation	
13		KNN implementation	
14		Naïve Bayes implementation	
15		Machine learning using scikit learn	
16		Feature selection using K-fold validation	
17-18		Final term	

Prepared by:

Date:

Approved by the Department on:
